OUR LADY OF FATIMA UNIVERSITY

COLLEGE OF PHARMACY

Valenzuela. Quezon City. Antipolo. Pampanga. Cabanatuan. Laguna

DRUGS FOR BRONCHIAL ASTHMA

PHARMACOLOGY 2

Name of Lecturer

Instructor- College of Pharmacy

COURSE FACILITATOR

Date

TIME

UNIT OUTCOMES

At the end of the lecture, the students should be able to:

- Understand the pathophysiology of bronchial asthma.
- Identify and describe the different types of bronchial asthma.
- Classify the different drugs used to relieve and control bronchial asthma.

WATCH THESE VIDEOS

(HowdoesAsthmaworks?)
https://www.youtube.com/watch?v=PzfLDi-sL3w

(ClinicalPresentationofAsthma)
https://www.youtube.com/watch?v=gvxF_TJhHiA

(How to use Metered Dose Inhaler?)
https://www.youtube.com/watch?v=S7OUiyXzABM

RESPIRATION

Refers to the movement of air in and out of the lungs through a series of air passages.
Nose
Mouth
Trachea
Bronchial tree

The branches of the bronchial tree are lined with **smooth muscle,** which adjusts the **constriction and dilation** of the airways in response to the **needs of the body.**

RESPIRATION

BODY SYSTEM AFFECTING RESPIRATION

1. Autonomic nervous system
regulate smooth muscle tone in the respiratory system and thereby maintain the balance between bronchoconstriction and bronchodilation

Paraympathetic :
- Bronchoconstriction
- Pulmonary vasoconstriction

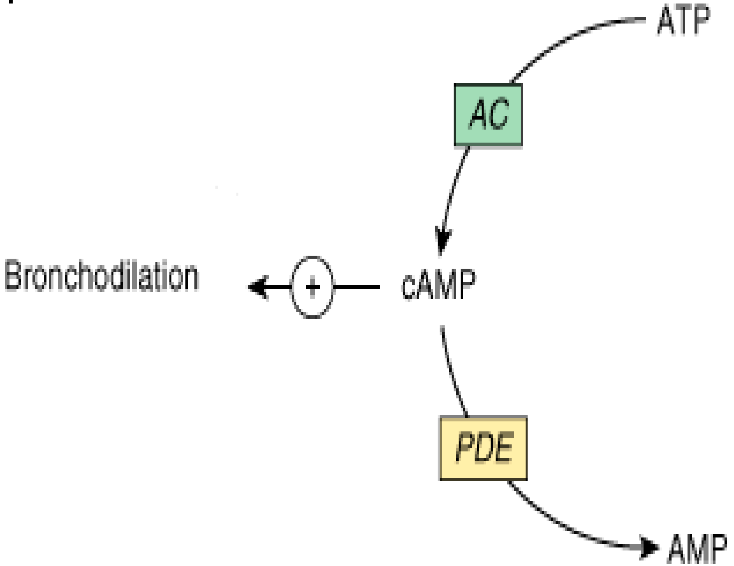
Sympathetic :
- Bronchodilation
- Pulmonary vasodilation

RESPIRATION

BODY SYSTEM AFFECTING RESPIRATION
2. cAMP

Smooth Muscle Relaxation

- Promotes bronchodilation
- Pulmonary Vasodilation



BRONCHIAL ASTHMA

- Physiologically characterized by increased responsiveness of the trachea and bronchi to various stimuli
- A recurrent, episodic shortness of breath
- Caused by **bronchoconstriction** from airway **inflammation and hyperreactivity.**

Caused by:
Genetic
Environmental factors

BRONCHIAL ASTHMA

Bronchial reactivity

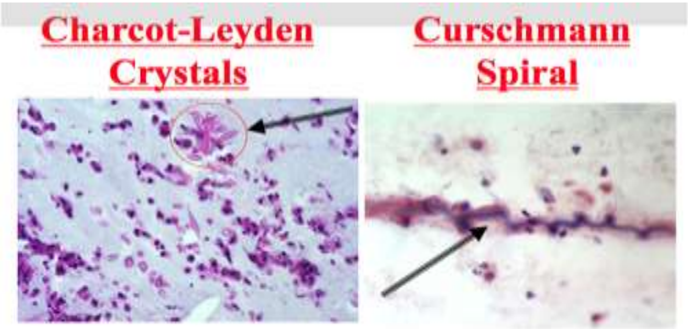
- is assessed by measuring the fall in forced expiratory volume in 1 second **(FEV1)**
- provoked by inhaling serially increasing concentrations of aerosolized **methacholine.**

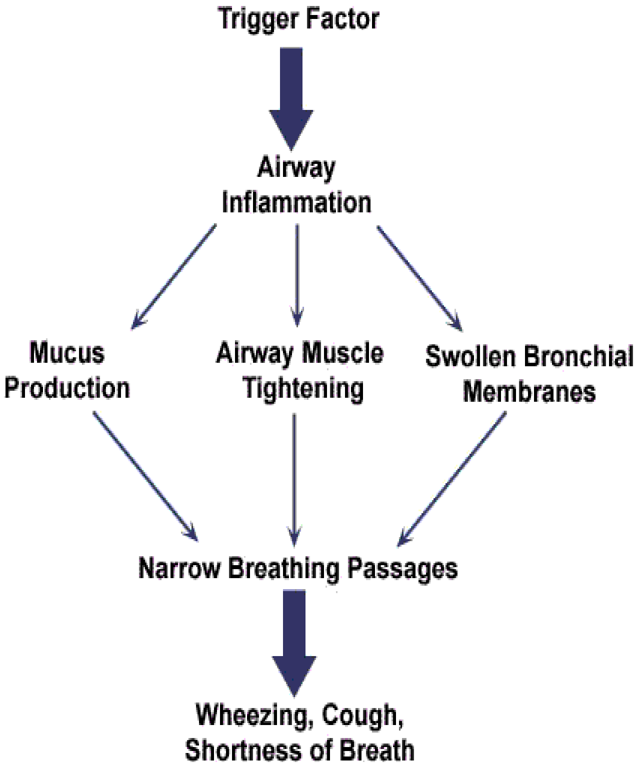
BRONCHIAL ASTHMA

- ❑ **Persistent cough**
- ❑ **Dyspnea**
- ❑ **Wheezing**
- ❑ **Tightness of the chest**
- ❑ **Sputum**

  Curschmann spiral

  Charcott Leyden crystals





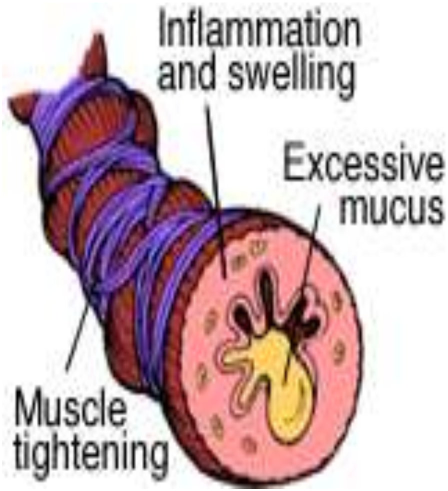
Trigger	Examples
Allergens 	Pollens, moulds, house dust mites, animals (dander, saliva, urine)
Industrial chemicals	Manufacture of, for example isocyanate-containing paints, epoxy resins, aluminum, hair sprays, penicilliins and cimetidine
Drugs 	Aspirin, Ibuprofen and other prostaglandin synthetase inhibitors, beta-adrenoceptor blockers
Food 	A rare cause but examples are nuts, fish, seafood, dairy products, food coloring, especially tartrazine, benzoic acid and sodium metabisulfite
Other industrial triggers 	Wood or grain dust, colophony, cotton dust, grain weevils and mites, also environmental pollutants such as cigarettes, cigarette smoke and sulfur dioxide
Miscellaneous	Cold air, exercise, hyperventilation, viral respiratory tract infections, emotion and stress.

PATHOPHYSIOLOGY OF ASTHMA

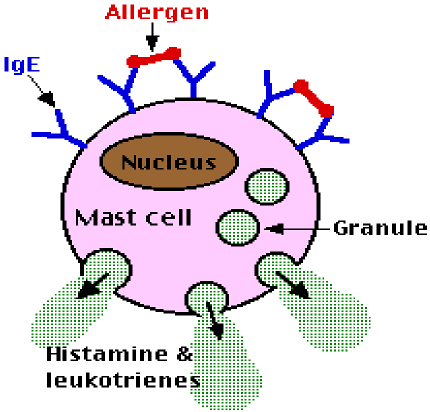
- First thing that happens during asthma is that there is **bronchospasm** which narrows the airways and 2nd there is increase **mucus secretion.**
- Which when combined can make the individual difficult to breath that's why asthma is a type obstructive pulmonary disease.

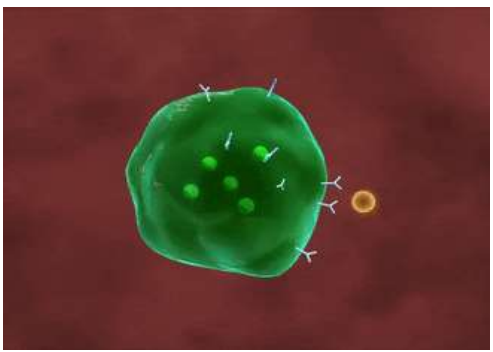
5 main events that occur in asthma

1. Triggering
2. Signalling
3. Migration
4. cell activation
5. tissue stimulation and damage



PATHOPHYSIOLOGY OF ASTHMA





1. TRIGGERING
Allergens
Non-allergic factors
aspirin, viral infections

- These antigen binds to IgE, which is attached to **activate mast cells**
- Activation of mast cells will result to release of histamine & leukotriene.

PATHOPHYSIOLOGY OF ASTHMA

2. SIGNALLING

- After triggering, mast cells and other signaling cells are activated (lymphocytes, eosinophils, epithelial cells, macrophages)

These inflammatory cells release chemical signals (cytokines, chemokines, eicosanoids, leukotrienes)

These chemical signals attract additional inflammatory cells to the airways

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Inflammatory Cells

EOSINOPHILS

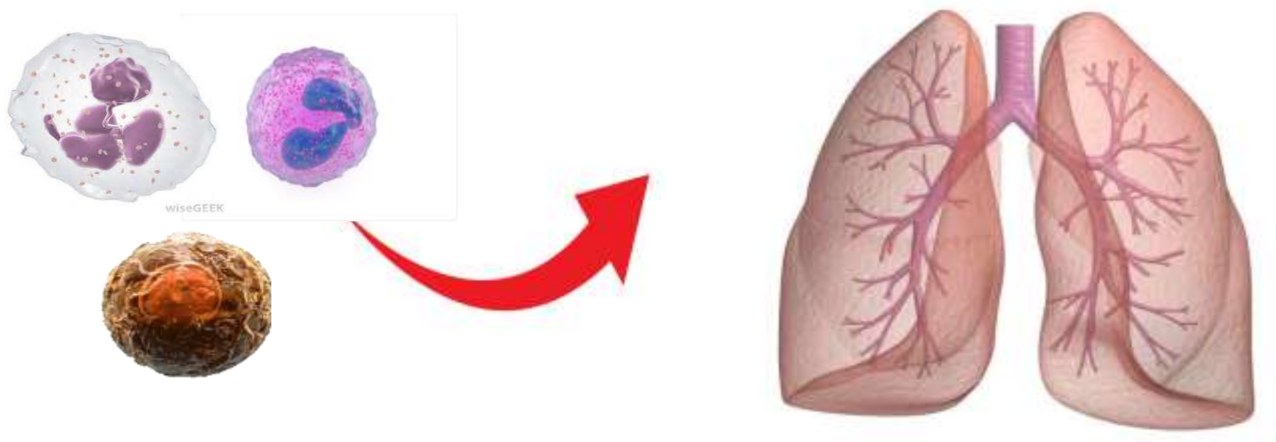
TYPE 2 HELPER
T CELL

NEUTROPHILS

PATHOPHYSIOLOGY OF ASTHMA

C. MIGRATION

- An influx of inflammatory cells (eosinophils, lymphocytes, monocytes, granulocytes)
- **Upregulation of adhesion** molecules begins
- These adhesion molecules affix themselves to cells in the circulation and attract these cells to the airways.



PATHOPHYSIOLOGY OF ASTHMA

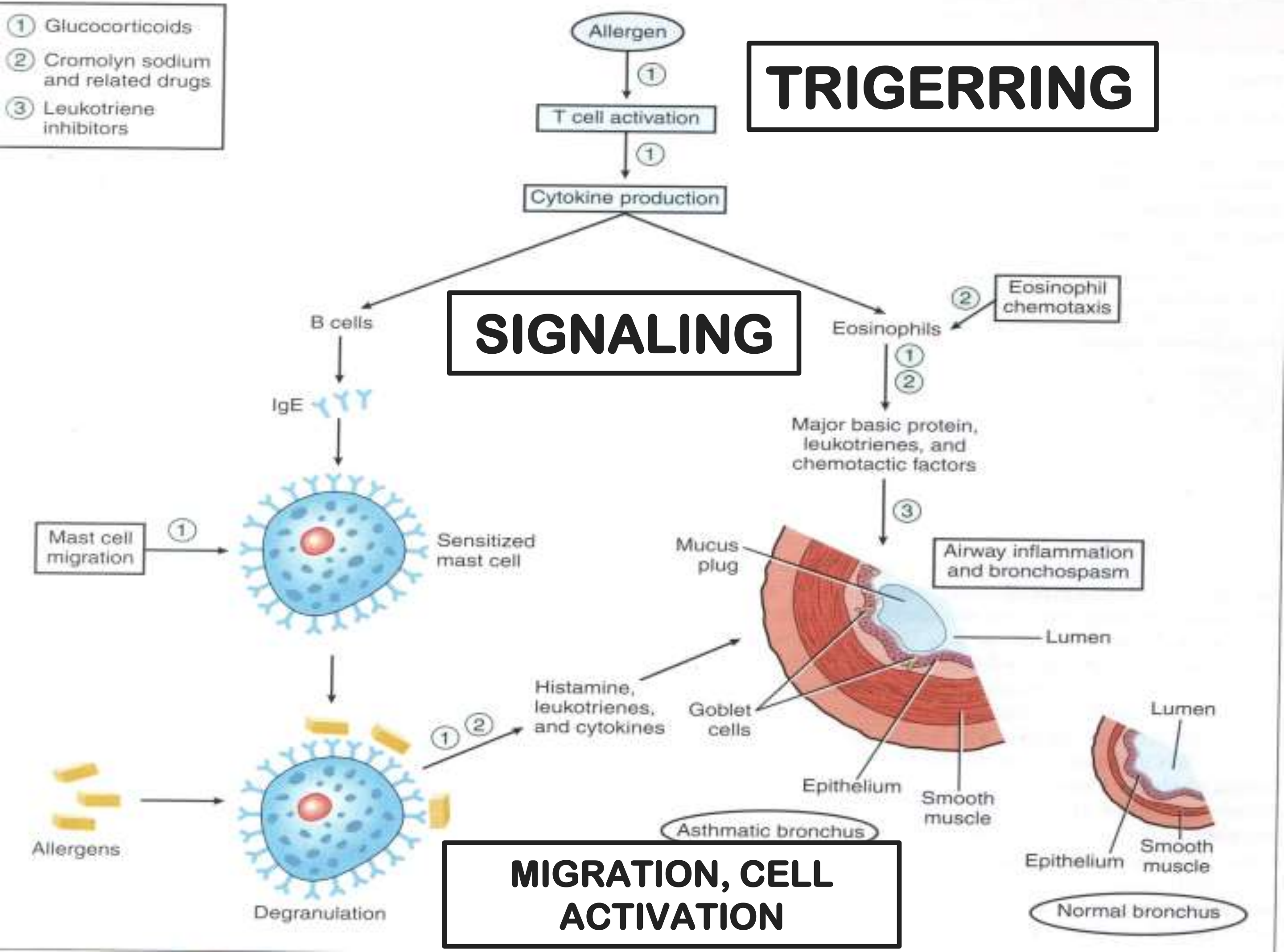
D. CELL ACTIVATION

- Required before cells can release inflammatory mediators. Once present in the airways, eosinophils are activated. **Leukotrienes & Cytokines appear to be important in this cell activation.**
- Once activated, they will **release inflammatory mediators.** These inflammatory mediators cause smooth-muscle constriction and set the stage for the late phase of the inflammatory process by initiating chemotaxis.

PATHOPHYSIOLOGY OF ASTHMA

Inflammatory Mediators:

- Histamine
- Leukotrienes
- Prostaglandins
- Bradykinin
- Adenosine
- other chemotactic agents that attract eosinophils and nuetrophils
- prostaglandin generating factor of anaphylaxis

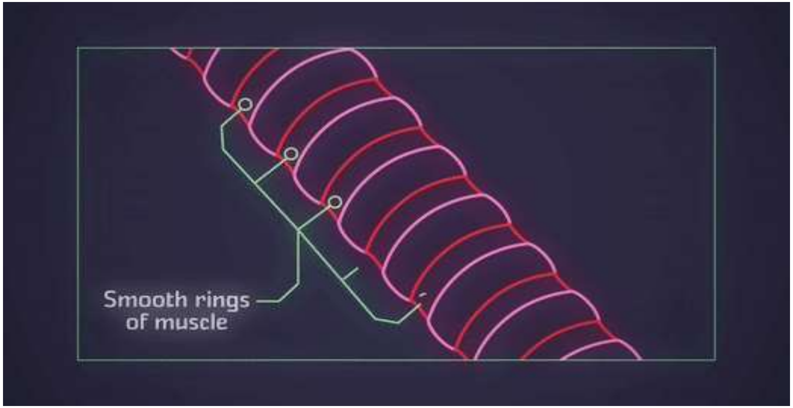


① Glucocorticoids
② Cromolyn sodium and related drugs
③ Leukotriene inhibitors
Allergen
①
T cell activation
①
Cytokine production
TRIGERRING
SIGNALING
B cells
IgE
Eosinophils
② Eosinophil chemotaxis
① ②
Major basic protein, leukotrienes, and chemotactic factors
③
Mast cell migration
①
Sensitized mast cell
Mucus plug
Airway inflammation and bronchospasm
Lumen
Histamine, leukotrienes, and cytokines
① ②
Goblet cells
Allergens
Epithelium
Smooth muscle
Asthmatic bronchus
Lumen
Epithelium
Smooth muscle
Normal bronchus
Degranulation
MIGRATION, CELL ACTIVATION

PATHOPHYSIOLOGY OF ASTHMA

E. TISSUE STIMULATION & DAMAGE

- occurs as a result of these inflammatory mediators released from activated cells.
- The epithelial damage is believed to contribute to airway hyperresponsiveness.
- Rather than resolving, airway inflammation may persist and result in a structural change to the lung known as **airway remodeling.**



TYPES OF ASTHMA

Determined by cause

1. EXTRINSIC ASTHMA

- Common in children
- Associated with a genetic predisposition
- Precipitated by known allergens

2. INTRINSIC ASTHMA (non-IgE mediated)

- unknown origin
- tends to develop in adulthood
- symptoms are triggered by non-allergenic factors.

TYPES OF ASTHMA

ASTHMA	No. of attacks	Nocturnal attacks
Intermittent asthma	Attacks occur in less than one per week	≤ 1 per month
Mild persistent asthma	attacks occur more than or once a week but not daily	≤ 1 per week
Moderate persistent	attacks occur daily but not more than once a day	≥ 1 a week
Severe persistent	attacks occur more than once a day	every night

TREATMENT GOALS

1. **REDUCE IMPAIRMENT**
2. **REDUCE RISK**

DRUGS FOR ASTHMA

“Short-term RELIEVERS”

- Bronchodilators
- These are agents that treat asthma attacks right away.
- agents that increase airway caliber by relaxing airway smooth muscle

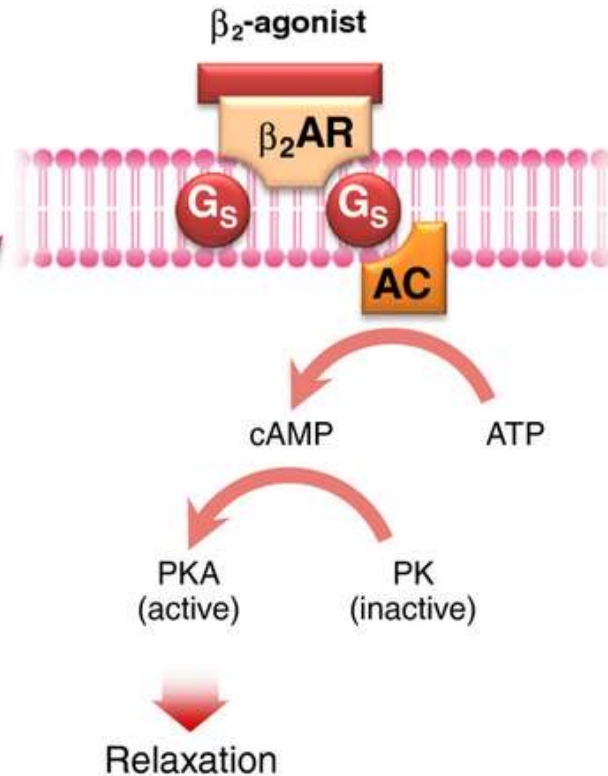
    β2 agonist
    Methylxanthines
    Anticholinergics/Antimuscarinics

“Long-term CONTROLLERS”

- Treat asthma over a long term (maintenance management)
- Anti-inflammatory agent such as

    inhaled corticosteroid
    Leukotriene antagonist
    Mast cell stabilizer

BETA-2-AGONISTS



Mechanism of Action:

- Stimulate ß2-receptors
- activating *adenyl cyclase*
- increases intracellular production of CAMP
- bronchodilation, improved mucociliary clearance, and reduced inflammatory cell mediator release

SABA- relievers
(acute asthma attacks)
LABA- controllers
(exacerbation of asthma)
(+ corticosteroids)

BETA-2-AGONISTS

OTHER EFFECTS:

Uterus:	Uterine muscle relaxation
Blood vessels:	Vasodilation
Skeletal muscle:	Tremor Stimulates Na/K ATPase Pump

BETA-2-AGONISTS

A. SHORT ACTING W/ RAPID ONSET

ex:

Salbutamol (Ventolin®)
Terbutaline (Bricanyl®)
Pirbuterol (Maxair®)
Fenoterol

- First line in the treatment of acute exacerbation of bronchial asthma
- Primary reliever medication
- Available as inhalers (Salbutamol and Terbutaline are also given as oral tablet)
- Terbutaline= tocolytic agents

BETA-2-AGONISTS

B. LONG ACTING W/ SLOW ONSET
ex:
Bambuterol (Bambec®)

- It is the only beta-2-agonist administered orally (without inhalational route)
- Prophylactic agent to reduce the frequency and severity of acute attacks.
- For controlling nocturnal attacks
- Controller medication

BETA-2-AGONISTS

C. LONG ACTING W/ RAPID ONSET

ex: Formoterol
Salmeterol (Serevent®)

- controller medication
- usually given through inhalation
- combined with corticosteroids

Salmeterol + Fluticasone (Seretide)
Formoterol + Budesonide (Symbicort)
Corticosteroids enhances the expression of beta 2 receptors.

BETA-2-AGONISTS

Adverse Effects:

- skeletal muscle tremor
- nervousness
- occasional weakness

Toxicities:

- Tachyphylaxis or tolerance *(occur with regular use of inhaled of oral ß-agonists)*
- Cardiac arrhythmias
- Hypoxemia

SYMPATHOMIMETICS

Epinephrine
Ephedrine
Isoproterenol

Bronchodilators
reserved for special situations (if cardiac stimulation is also needed)

SYMPATHOMIMETICS

EPINEPHRINE

- Stimulates catecholamine receptor including beta 2
- is an effective, rapidly acting bronchodilator
- SQ: subcutaneously (0.4 mL of 1:1000 solution)
- INH: inhaled as a micro-aerosol from a pressurized canister (320 g per puff).
- Can cause tachycardia, arrhythmias, and worsening of angina pectoris due to stimulation of beta 1 receptor.
- Use: Anaphylactic shock

SYMPATHOMIMETICS

EPHEDRINE

- ❑ An alkaloid from Ephedra sinica
- ❑ Increase catecholamine release
- ❑ Inhibition of reuptake
- ❑ Compared to epinephrine, ephedrine has:
  - longer duration
  - Administered orally
  - more pronounced central effects
  - lower potency

SYMPATHOMIMETICS

ISOPROTERENOL

- Isoprenaline®
- A potent bronchodilator
- It is a no—selective beta agonist
- when inhaled as a microaerosol from a pressurized canister, 80–120 g causes maximal bronchodilation within 5 minutes.
- It has a 60- to 90-minute duration of action.

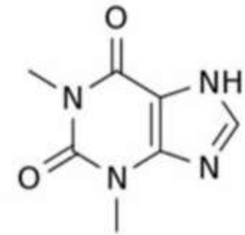
METHYLXANTHINES

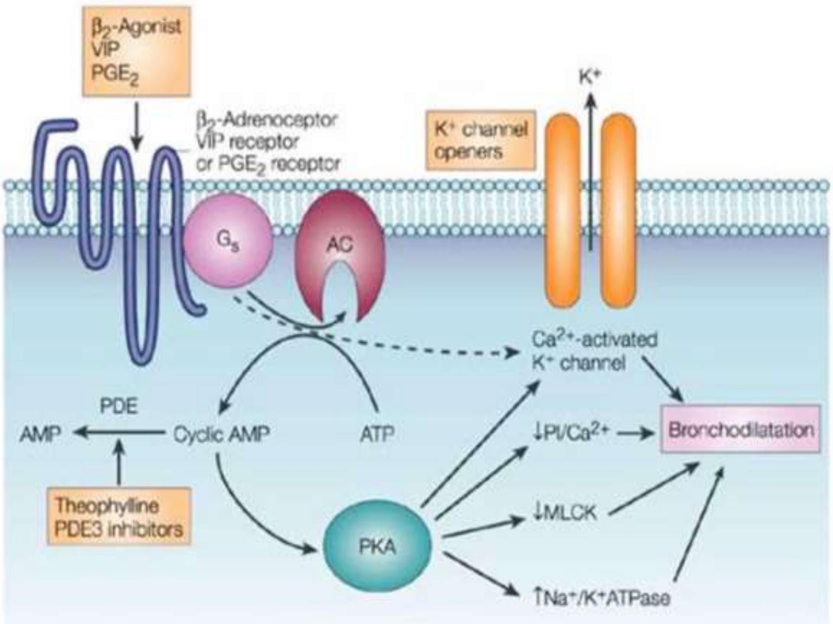
Alternative controller medication because it possesses narrow therapeutic index

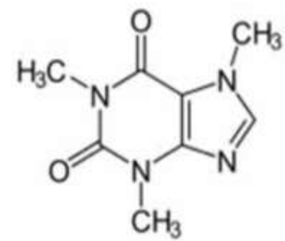
Mechanisms of Action:

- ❑ At high concentrations, it inhibit several members of the phosphodiesterase (PDE) enzyme family
- ❑ At low doses, they inhibit the late phase of allergic reaction and may they may actually have anti inflammatory activity.
- ❑ Inhibition of cell surface receptors for adenosine.

Theophylline







caffeine

PDE=Phosphodiesterase

PKA=Protein Kinase A

METHYLXANTHINES

Examples:

- Theophylline
- Aminophylline (80% theophylline)- a theophylline-ethylenediamine complex
- Doxofylline
- Dyphylline- a synthetic analog of theophylline less potent and shorter-acting than theophylline
- Phentoxyfylline (not for asthma-for PAD/intermittent claudication)

METHYLXANTHINES

Clinical uses:

- Alternative controller medication for asthma
- Prevent nocturnal attacks
- Sustained release formulation are given at bedtime
- For the treatment of COPD (usually given at bedtime)
- Alternative reliever for asthma

METHYLXANTHINES

A major adverse effect of the PDE4-selective drugs is **nausea and vomiting.**

Higher levels (> 40 mg/L) may **cause seizures or arrhythmias**

- CNS – stimulation, anxiety, irritability states, seizures
- Cardiovascular – tachycardia, palpitations, arrhythmia
- GIT- stimulate secretion of both gastric acid and digestive enzymes.
- Diuretic effect- involve both increased glomerular filtration and reduced tubular sodium reabsorption

ANTICHOLINERGICS

Ipratropium
Tiotropium- With 24-hour duration of action (Long acting)
Oxytropium

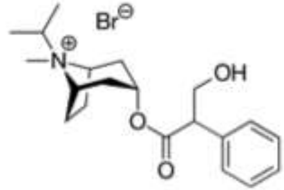
IPRATROPIUM (Atrovent®)

- Datura stramonium/
- Datura metel

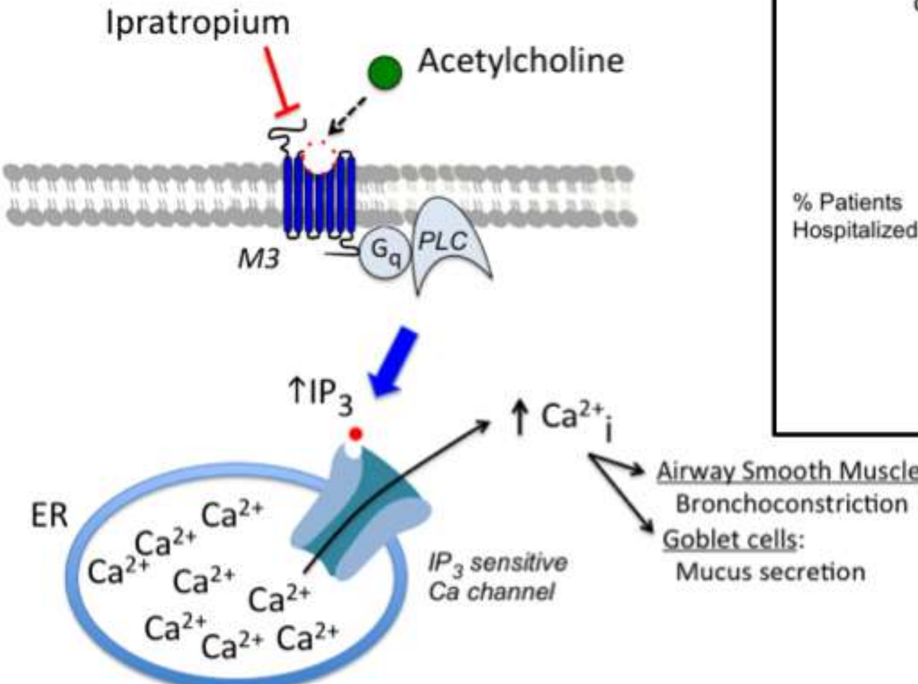
MOA:

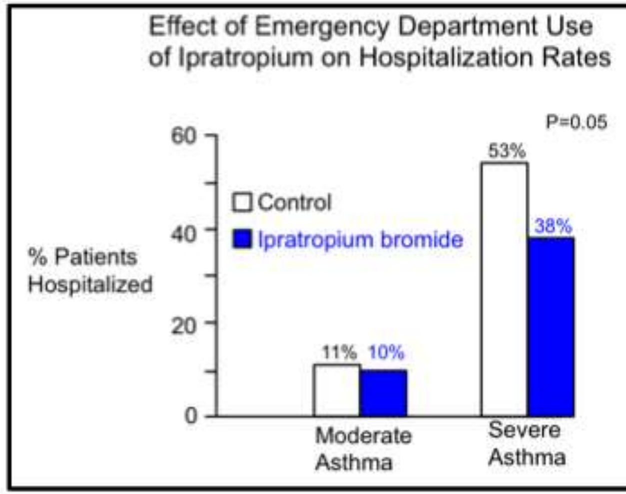
- competitively inhibit the effect of acetylcholine at muscarinic receptors thus blocking the contraction of the airway smooth muscle.

Ipratropium Bromide



Airway smooth muscle & mucous cells





Qureshi F et al.
NEJM 339:1030, 1998

ANTICHOLINERGICS

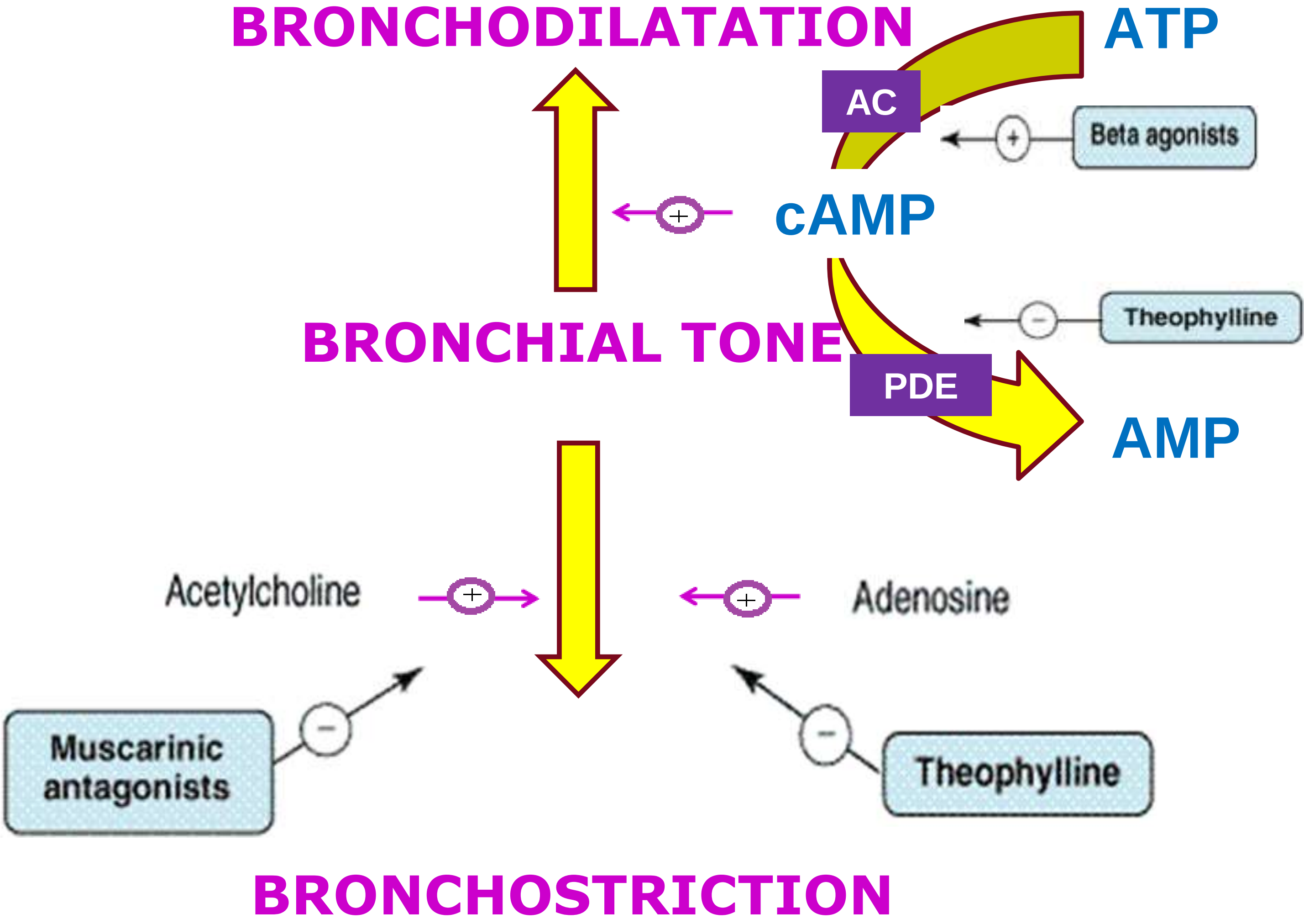
Clinical Characteristics:

- more useful in COPD than in asthma
- has minimal systemic toxicity-no atropine like effects
- has a better safety profile compared to β2 agonist
- slightly less effective than beta-agonist agents in reversing asthmatic bronchospasm
- the addition of ipratropium enhances the bronchodilation produced by nebulized albuterol in acute severe asthma.

Ipratropium (Atrovent)- more peripheral effects (LUNGS) than CNS

Ipratropium + Fenoterol (Berodual)

Ipratropium + Salbuterol (Combivent)



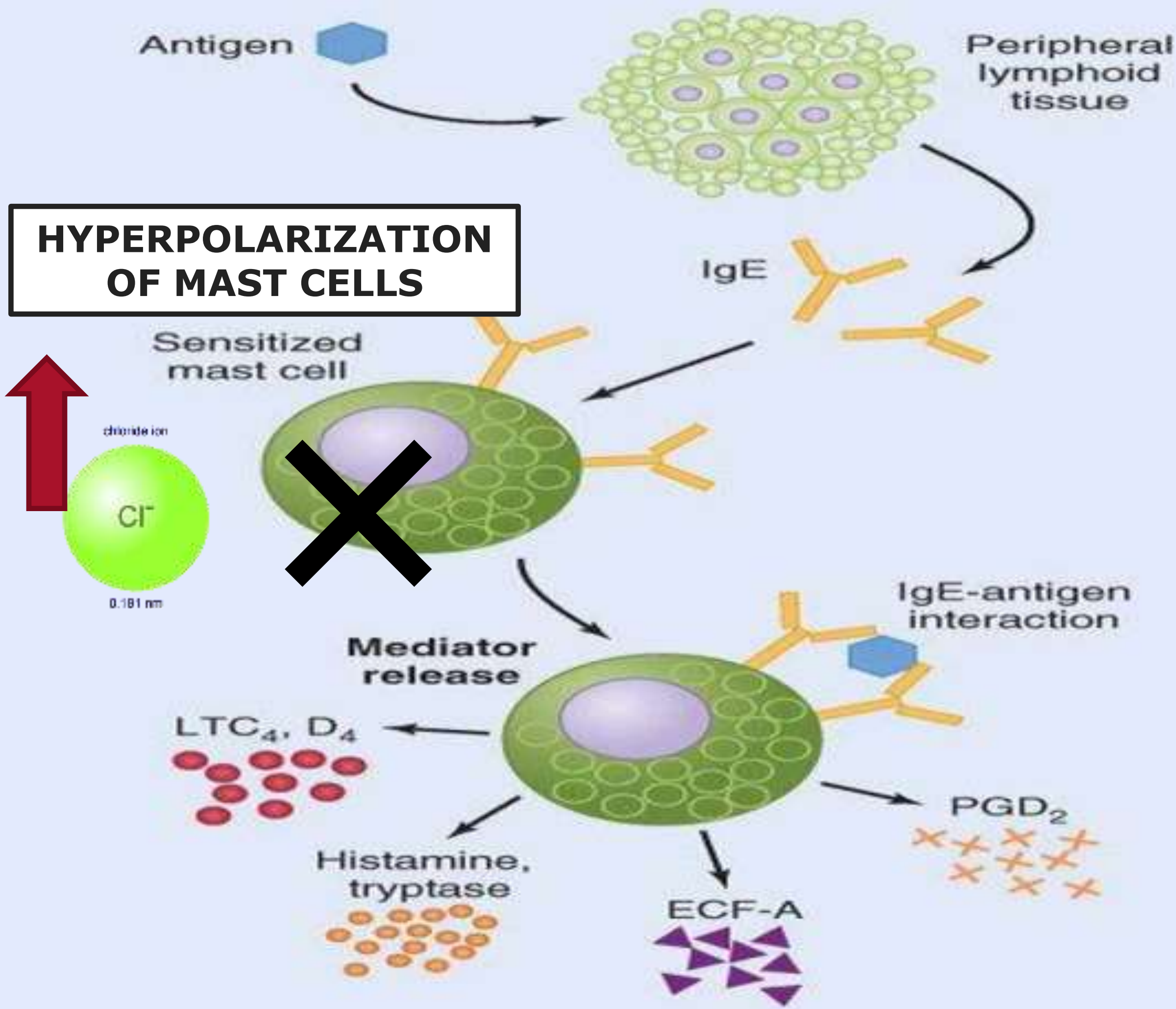
BRONCHODILATATION
ATP
AC
Beta agonists
cAMP
Theophylline
BRONCHIAL TONE
PDE
AMP
Acetylcholine
Adenosine
Muscarinic antagonists
Theophylline
BRONCHOSTRICTION

MAST CELL STABILIZERS

- **Prophylactic agents for asthma**
- It is stable but extremely insoluble salts
- When used as aerosols (metered-dose inhalers), they effectively inhibit both antigen- and exercise-induced asthma, and chronic use (four times daily) slightly reduces the overall level of bronchial reactivity.

Mechanism of action:
stabilizes the membrane of the mast cells by increasing inward conduction of membrane to chloride ions thus inhibiting cellular activation.

HYPERPOLARIZATION OF MAST CELLS



MAST CELL STABILIZERS

CROMOLYN (Intal®)

- poorly absorbed from the gastrointestinal tract and must be inhaled as a microfine powder or aerosolized solution.

NEDOCROMIL (Tilade®)

- has a very low bioavailability and is available only in metered-dose aerosol form.

Clinical uses:

1. Given for asthma as inhalation
2. May also be useful in allergic rhinitis
3. Cromolyn solution is also useful in reducing symptoms of allergic rhinoconjunctivitis.
4. used as prophylaxis of EIB

MAST CELL STABILIZERS

SIDE EFFECTS:

- Bronchospasm (acute administration)

Because the drugs are so poorly absorbed, adverse effects of cromolyn and nedocromil are minor and are localized to the sites of deposition.

These include:

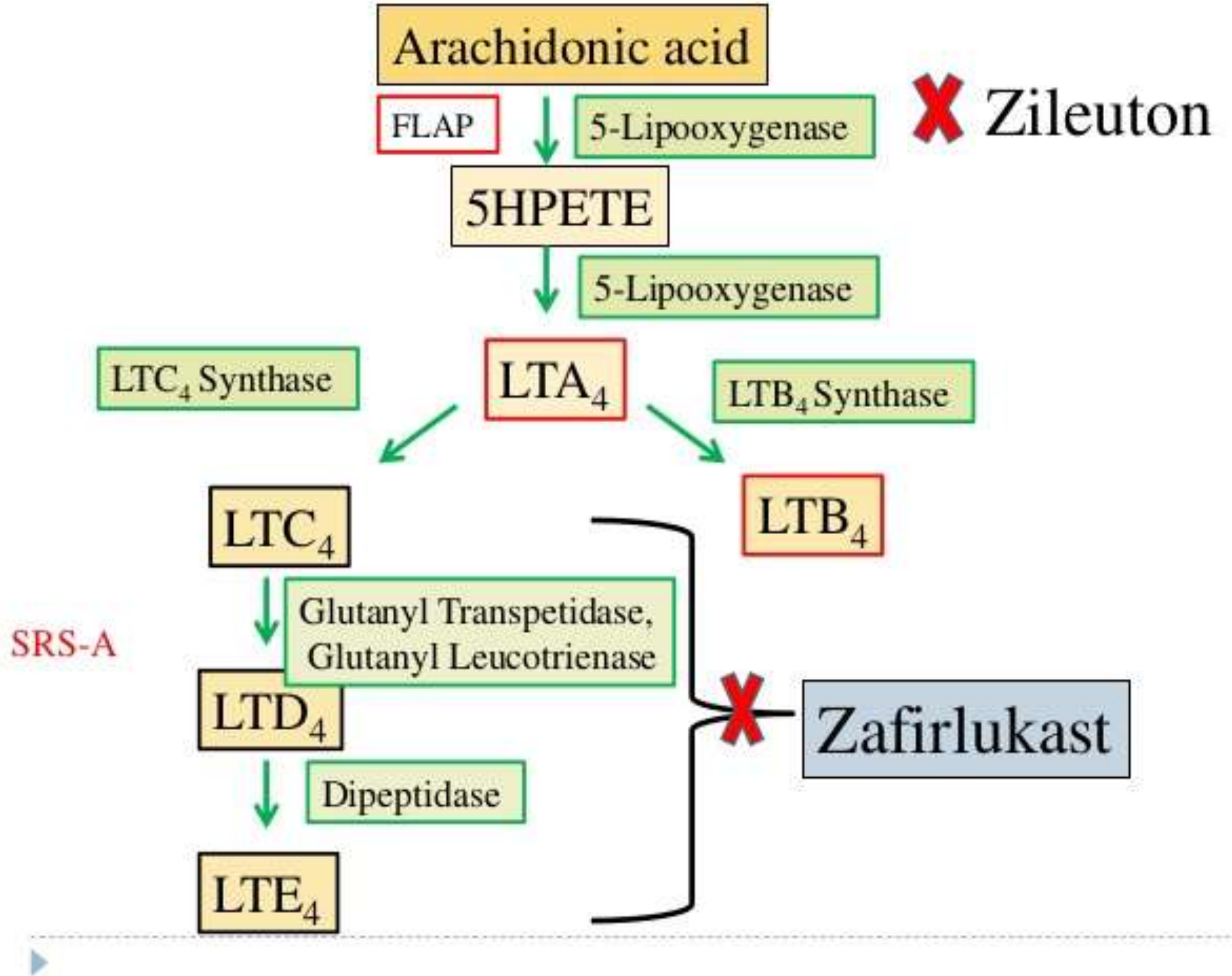
- throat irritation
- Cough
- mouth dryness
- chest tightness & wheezing
- Reversible dermatitis, myositis, or gastroenteritis

ANTI-INFLAMMATORY

Controller medication for asthma

A. LEUKOTRIENE MODIFIERS
   - LIPOXYGENASE INHIBITOR
   - LTD4 ANTAGONISTS

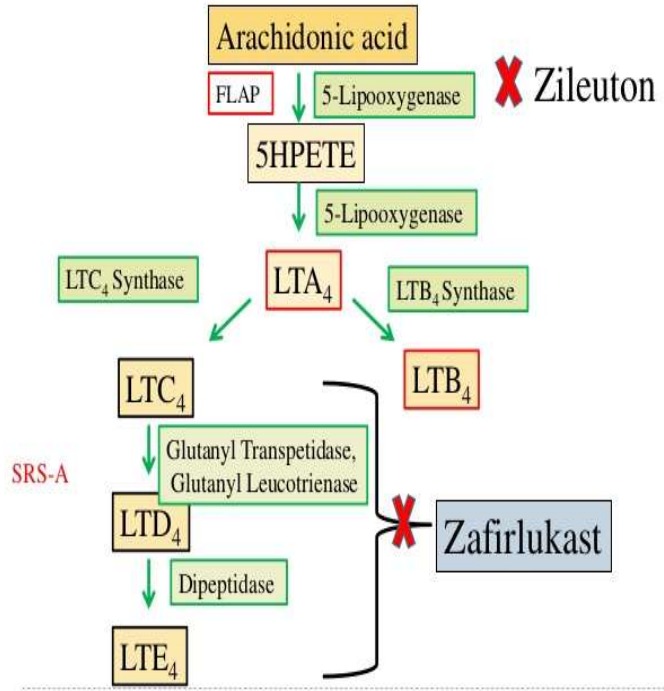
B. CORTICOSTEROIDS


Arachidonic acid
FLAP
5-Lipooxygenase
Zileuton
5HPETE
5-Lipooxygenase
LTC$_4$ Synthase
LTA$_4$
LTB$_4$ Synthase
LTC$_4$
LTB$_4$
SRS-A
Glutanyl Transpetidase,
Glutanyl Leucotrienase
LTD$_4$
Zafirlukast
Dipeptidase
LTE$_4$

LIPOXYGENASE INHIBITOR

Ex: Zileuton (Zyflo®)
is the least prescribed because of the requirement of four times daily dosing and because of occasional liver toxicity.

Mechanism of action:
inhibits 5-lipoxygenase thereby preventing leukotriene synthesis



LIPOXYGENASE INHIBITOR

Administration:

- In adults and children 12 years of age and older, the dosage of zilueton is 600 mg four times daily.
- It may be taken without regard to meals.

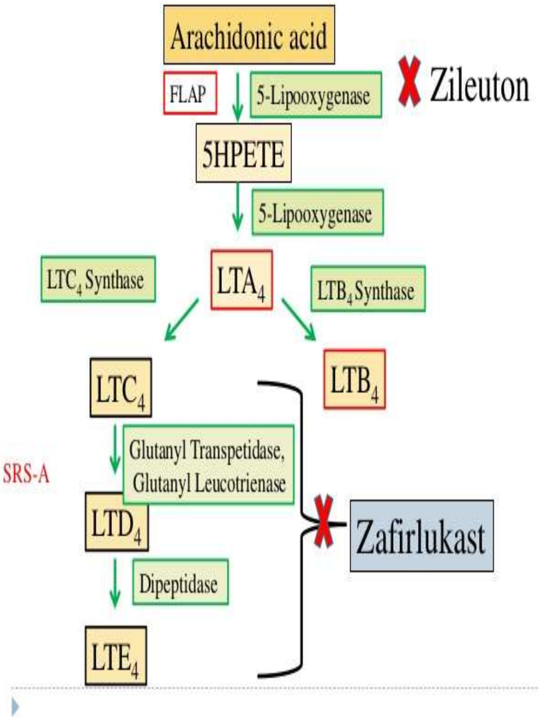
Contraindication:

- Zileuton is contraindicated in patients with hepatic function impairment and should be monitored closely in patients who drink large quantities of alcohol.

Adverse Effects:

- Adverse effects include headache, abdominal pain, asthenia, nausea, dyspepsia and myalgia.

LEUKOTRIENE ANTAGONISTS



Montelukast (Singulair®)
Zafirlukast (Accolate®)
MOA:

- prevents the binding of LTD4 to its specific receptor thus inhibiting its effects on airways.
- It is a selective cysteinyl leukotriene 1 (CysLT1) receptor antagonists; therefore, they prevent leukotrienes from interacting with their receptors.

LEUKOTRIENE ANTAGONISTS

Clinical uses:

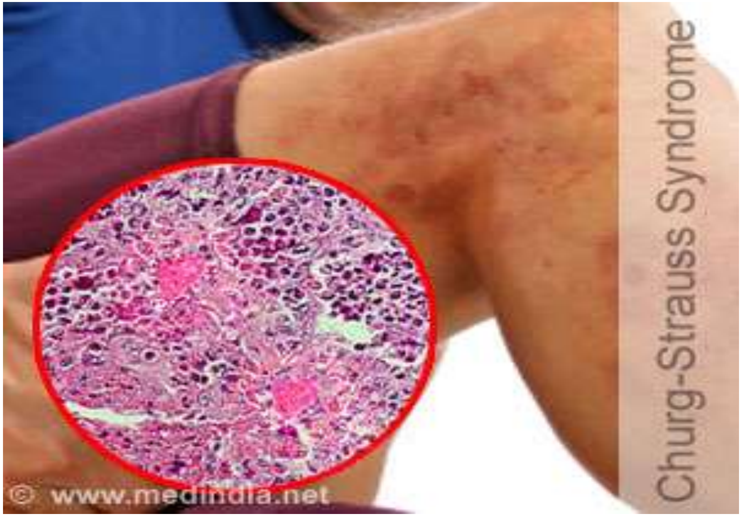
- NSAID induced asthma
- alternative controller for persistent bronchial asthma
- Montelukast- once a day dosing (beneficial, promote patient compliance)
- Zafirlukast- twice a day dosing

MONTELUKAST

- approved for children as young as 6 years of age.
- PRECAUTION: The chewable forms of montelukast (4- and 5-mg tablets) contain aspartame and should be avoided in patients with **phenylketonuria.**

LEUKOTRIENE ANTAGONISTS

SIDE EFFECT:
Mask the symptom of Churg-Strauss syndrome- a systemic vasculitis characterized by worsening asthma, pulmonary infiltrates, and eosinophilia



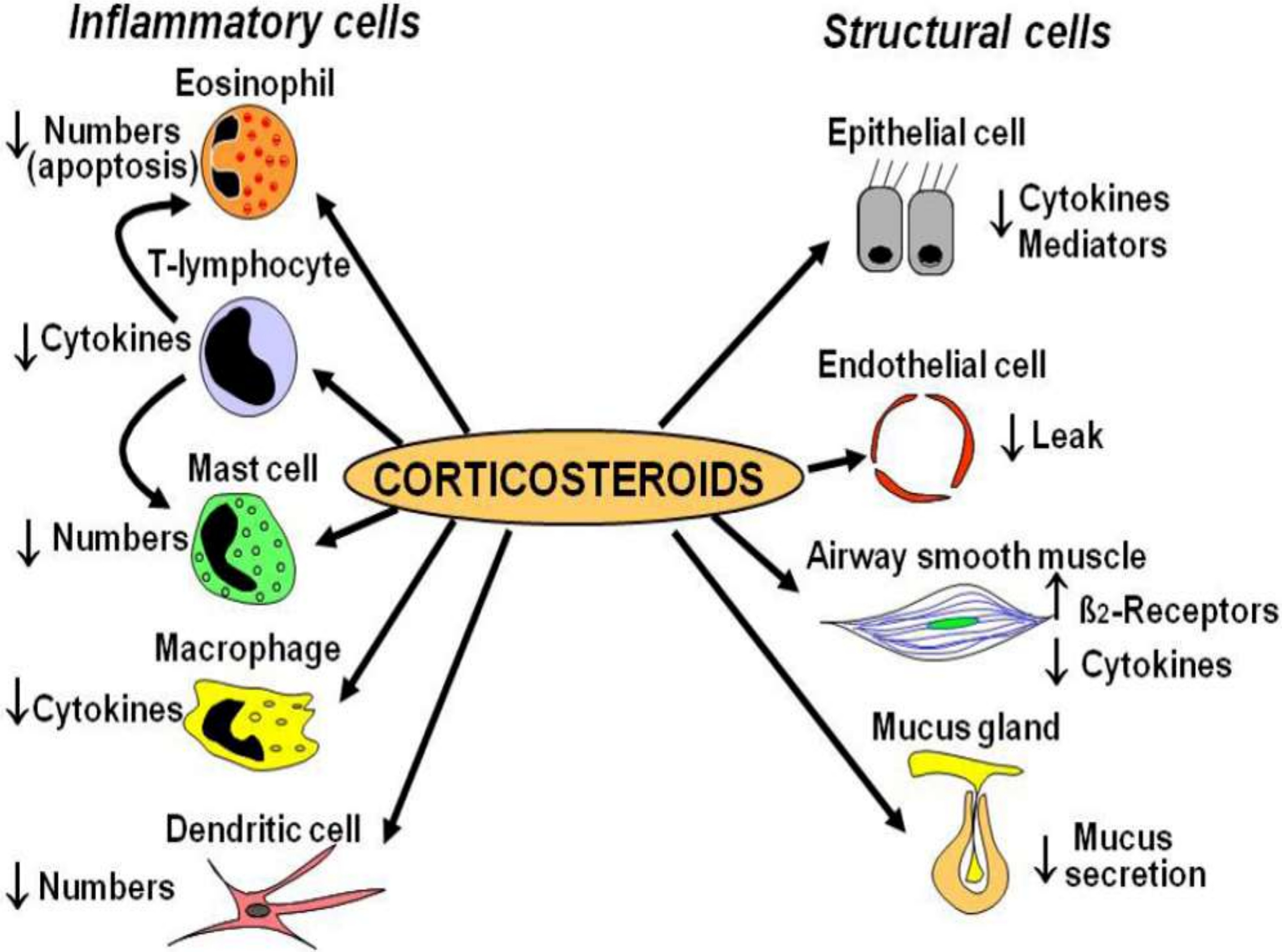
CORTICOSTEROIDS

MECHANISM OF ACTION:

- Corticosteroids bind to glucocorticoid receptors on the cytoplasm of cells.
- The activated receptor regulates transcription of target genes.

Corticosteroids reduce inflammation via:

- Inhibition of transcription and release of inflammatory genes.
- Increased transcription of anti-inflammatory genes that produce proteins that participate in or suppress the inflammatory process.


Inflammatory cells
Structural cells
Eosinophil
↓ Numbers (apoptosis)
T-lymphocyte
↓ Cytokines
Mast cell
↓ Numbers
Macrophage
↓ Cytokines
Dendritic cell
↓ Numbers
CORTICOSTEROIDS
Epithelial cell
↓ Cytokines Mediators
Endothelial cell
↓ Leak
Airway smooth muscle
↑ ß2-Receptors
↓ Cytokines
Mucus gland
↓ Mucus secretion

CORTICOSTEROIDS

GLUCOCORTICOIDS
A. LONG ACTING (inhaled)

Budesonide
Beclomethasone
Fluticasone
Flunisolide

- first line controller medication
- with minimal systemic side effect
- oral trush (candidiasis) vocal cord irritation (Gargle 3-4x after use)

B. SYSTEMIC ORAL MEDICATIONS

Prednisone/Prednisolone

- Short acting agents
- used in the management of severe asthma attacks/exacerbations

CORTICOSTEROIDS

C. SYSTEMIC PARENTERAL

Hydrocortisone
Methylprednisolone
- preferred agent for status asthmaticus

Status asthmaticus

- is a life-threatening condition that occurs when a severe asthma exacerbation fails to respond to usual treatment.

CORTICOSTEROIDS

Administration:

For control of asthma

- administer corticosteroids early in the morning, after endogenous ACTH secretion has peaked.
- because secretion of corticosteroids has a diurnal variation

For prevention of nocturnal asthma

- oral or inhaled corticosteroids are most effective when given in the late afternoon

AEROSOL TREATMENT

- is the most effective way to decrease the systemic adverse effects of corticosteroid therapy.

CORTICOSTEROIDS

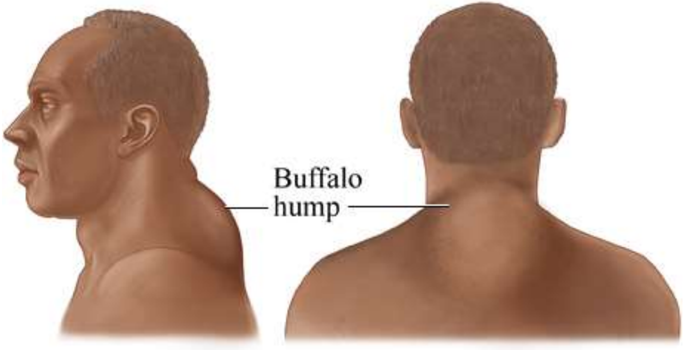
SIDE EFFECTS: inhaled corticosteroids can cause:

1. Oropharyngeal candidiasis & Hoarseness can also result from a direct local effect of inhaled corticosteroids on the vocal cords.
2. short-term complications in adults but may increase the risks of osteoporosis
3. cataracts over the long term.
4. In children, inhaled corticosteroid therapy (slow the rate of growth)

CORTICOSTEROIDS

SIDE EFFECTS: ORAL/ SYSTEMIC

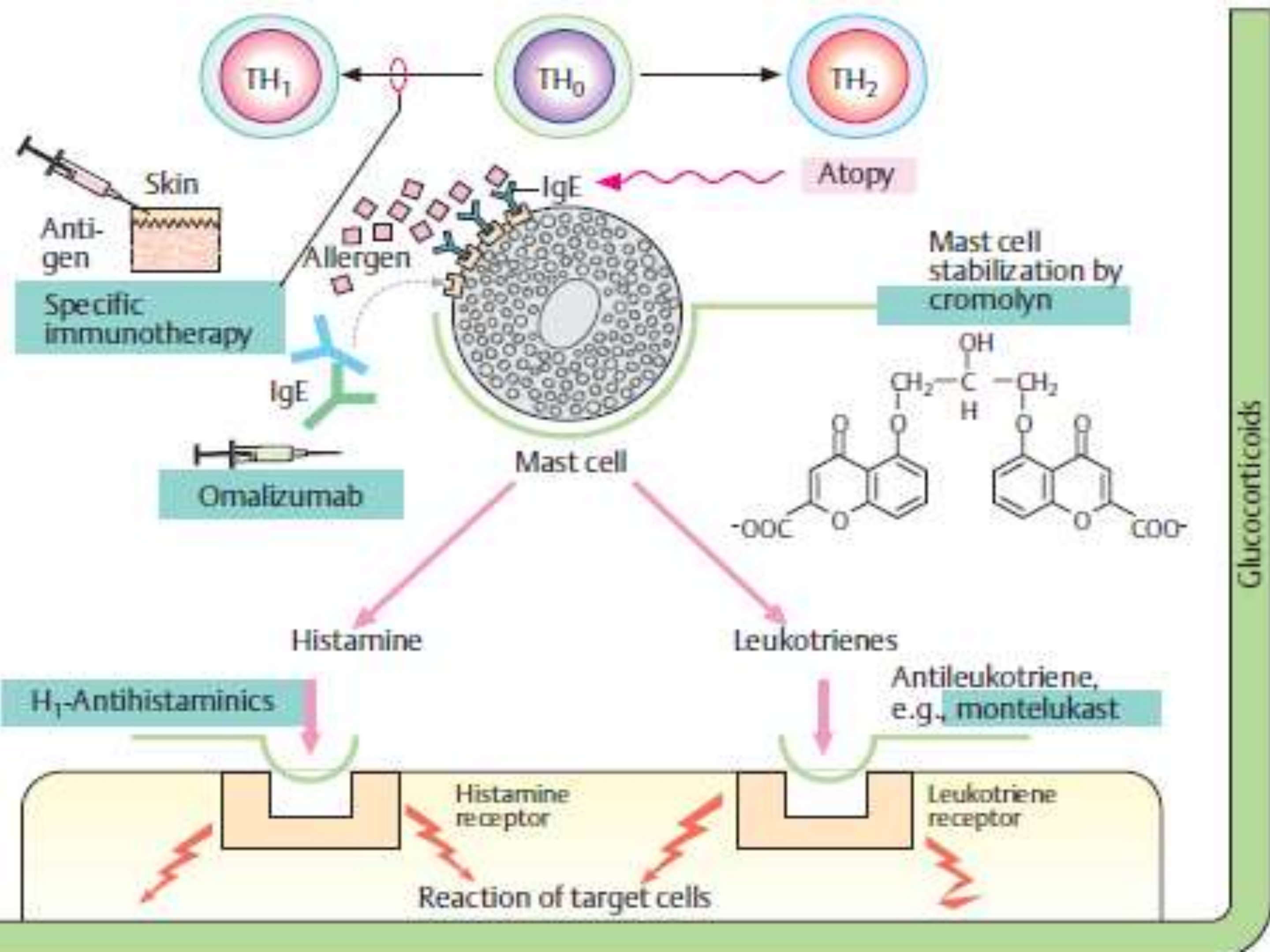
-Osteoporosis
-Cushing-like syndrome
-Cataracts
-Hyperglycemia



ANTI-IGE ANTIBODIES

Anti-IgE Monoclonal Antibodies
Ex: Omalizumab (anti-IgE Mab)

- inhibits the binding of IgE to mast cells
- but does not activate IgE already bound to these cells and thus does not provoke mast cell degranulation.



TH1
TH0
TH2
Skin
Anti-
gen
Specific
immunotherapy
IgE
Atopy
Allergen
Mast cell
stabilization by
cromolyn
IgE
Omalizumab
Mast cell
OH
CH2
C
CH2
H
O
O
O
-OOC
O
O
COO-
Glucocorticoids
Histamine
Leukotrienes
H1-Antihistaminics
Antileukotriene,
e.g., montelukast
Histamine
receptor
Leukotriene
receptor
Reaction of target cells

DRUG DELIVERY

Metered Dose Inhalers (MDIs)

- AZMACORT (triamcinolone) inhaler
- Breath-actuated MDIs
- require the patient to use a closed-mouth technique.

Remove cap and shake inhaler.

Breathe out all the way.

Breathe in and press down on the inhaler.

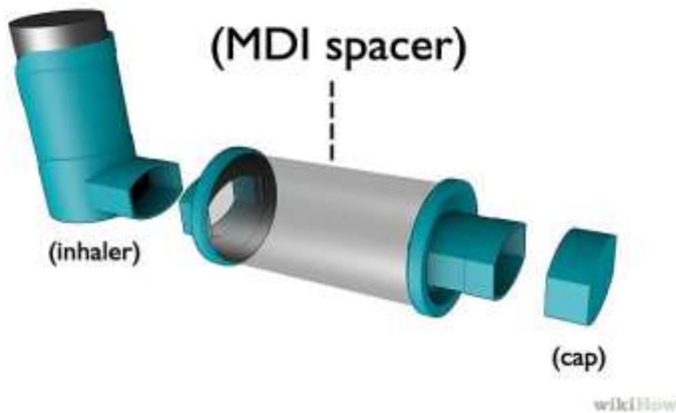
Hold your breath for a count of 10.

Slowly breathe out.

DRUG DELIVERY

SPACERS AND HOLDING CHAMBERS

- Additions of a spacer in a patient with poor MDI technique improves pulmonary delivery of the agent.
- (e.g., Aerochamber, Aerosol Cloud Enhancer, AeroVent, Brethancer, Ellipse, E-Z Spacer, Inhal-Aid, InspirEAse, OptiChamber, OptiHaler)



DRUG DELIVERY

NEBULIZERS

require less patient coordination during administration of multiple inhalations.

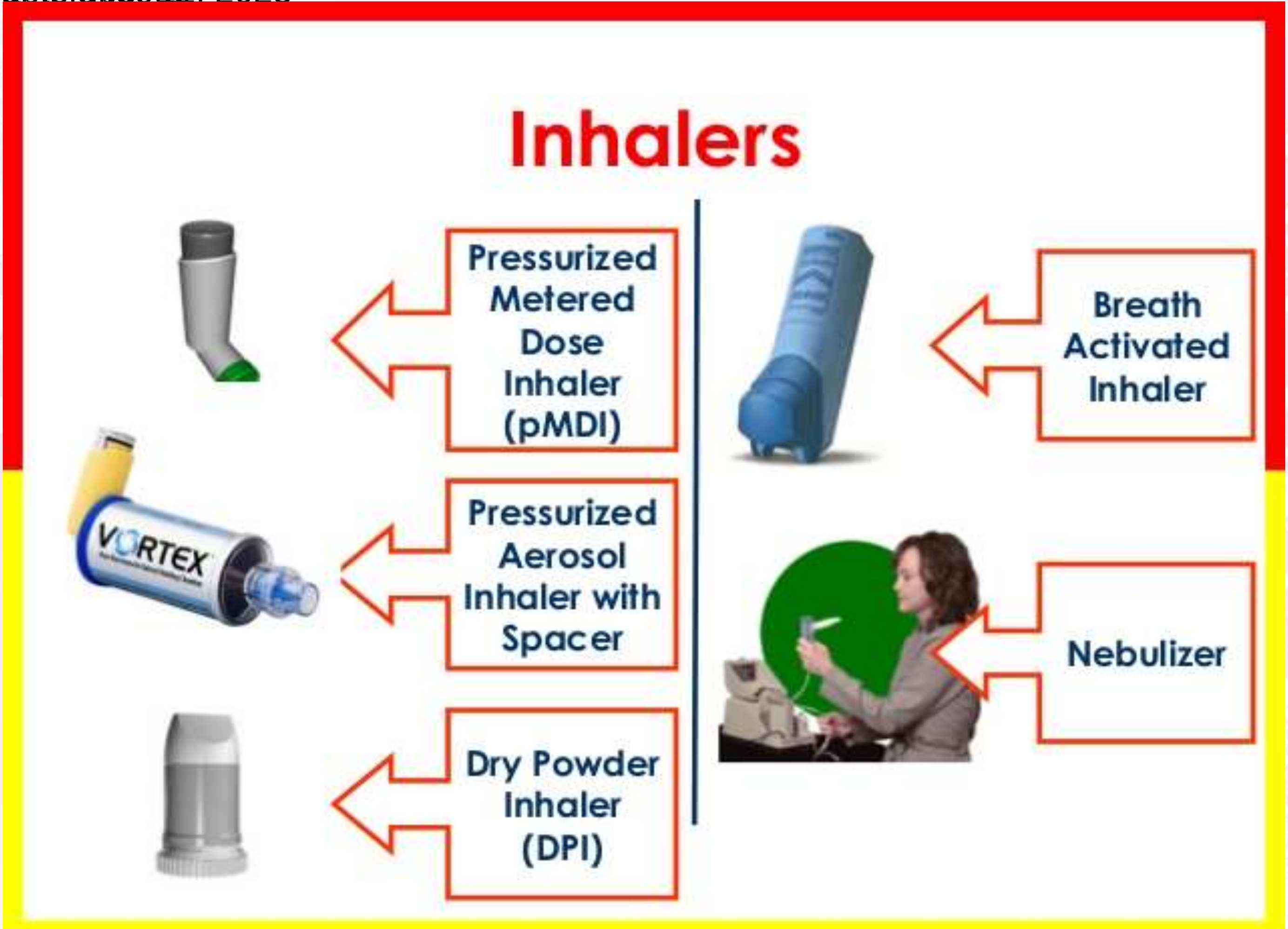
Disadvantages:

1) Cost
2) preparation and administration time
3) size of the device
4) drug delivery inconsistencies between devices.

DRUG DELIVERY

DRY-POWDER INHALERS

- to avoid the use of Freon propellants.
- They are also being used more frequently because many patients find them easier to use than an MDI.


Inhalers
Pressurized Metered Dose Inhaler (pMDI)
VORTEX
Pressurized Aerosol Inhaler with Spacer
Dry Powder Inhaler (DPI)
Breath Activated Inhaler
Nebulizer

NON-PHARMACOLOGIC

A. HUMIDIFIED OXYGEN

- Although the fraction of inspire oxygen (FIO2) administered is based on the patient’s arterial blood gas status, **1-3 L/min is generally given via facemask or nasal cannula.**
- The goal is to keep the SaO2 greater than 90% (greater than 95% if the patient is pregnant or has heart disease).

NON-PHARMACOLOGIC

B. HELIOX

- A mixture of **helium and oxygen** that has a lower density than air.
- Because of its decreased airflow resistance, heliox may increase ventilation during acute asthma exacerbations.

C. INTRAVENOUS FLUIDS

- and electrolytes may be required if the patient is volume depleted.

NON-PHARMACOLOGIC

D. ENVIRONMENTAL CONTROL

- Some measures include use of allergen-resistant mattress and pillow encasements, use of high-filtration vacuum cleaners, removal of carpets and draperies, and avoidance of furry pets.

E. VACCINES

- (e.g., influenze virus, polyvalent pneumococcals) are recommended to prevent infection, which may precipitate an exacerbation.

COMPLICATIONS

STATUS ASTHMATICUS
is a life-threatening condition that occurs when a severe asthma exacerbation fails to respond to usual treatment.

Standard therapy for status asthmaticus involves:

- oxygen administration, intravenous rehydration, inhaled β-agonists and anticholinergics, and intravenous corticosteroids.

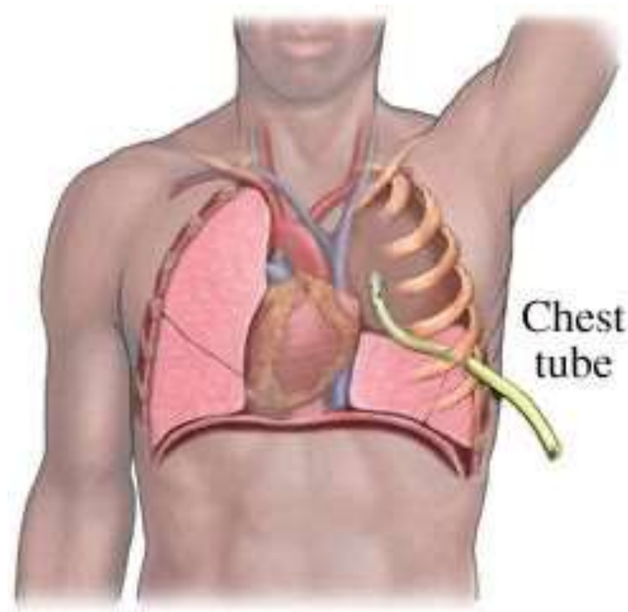
- Aggressive therapy is indicated for any patient with acute severe exacerbation and for patients who fail to respond to conventional therapy for acute exacerbation

COMPLICATIONS

PNEUMOTHORAX

- is a condition characterized by accumulation of air in the pleural space, as sometimes occurs during an acute asthma exacerbation.

- Therapy includes oxygen, aspiration of pleural air via a chest tube, and analgesics

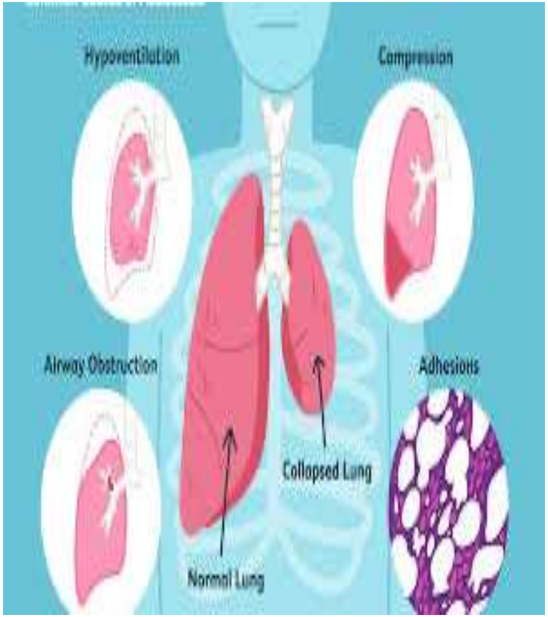


COMPLICATIONS

Atelectasis

- inhibits gas exchange during respiration
- may occur as a result of airway obstruction.
- In asthmatics, atelectasis usually involves the right middle lobe, but sometimes affects the entire lung.

Symptoms include **worsening dyspnea and anxiety.**



COMPLICATIONS

Physical findings include hyperventilation, diminished breath sounds, decreased chest excursion, mediastinal shift toward the affected side and cyanosis.

Therapy includes:

- Incentive spirometry
- Postural drainage
- Chest percussion
- Coughing and deep breathing exercises, and bronchodilators.
- Bronchoscopy

DRUGS FOR RESPIRATORY DISEASES

PHARMACOLOGY 2

zoom

Date

Name of Lecturer
Instructor- College of Pharmacy
COURSE FACILITATOR

OUTCOMES

At the end of this unit, the students are expected to:

- Understand the pathophysiology of different respiratory disorders.
- Identify and describe the different types of chronic obstructive pulmonary disease.
- Classify the different drugs used to manage respiratory disorders.

CHECKLIST

- Read course and unit objectives
- Read study guide prior to class attendance
- Read required learning resources; refer to unit terminologies for jargons
- Proactively participate in discussions
- Participate in discussion board (Canvas) ◉
Answer and submit course unit tasks

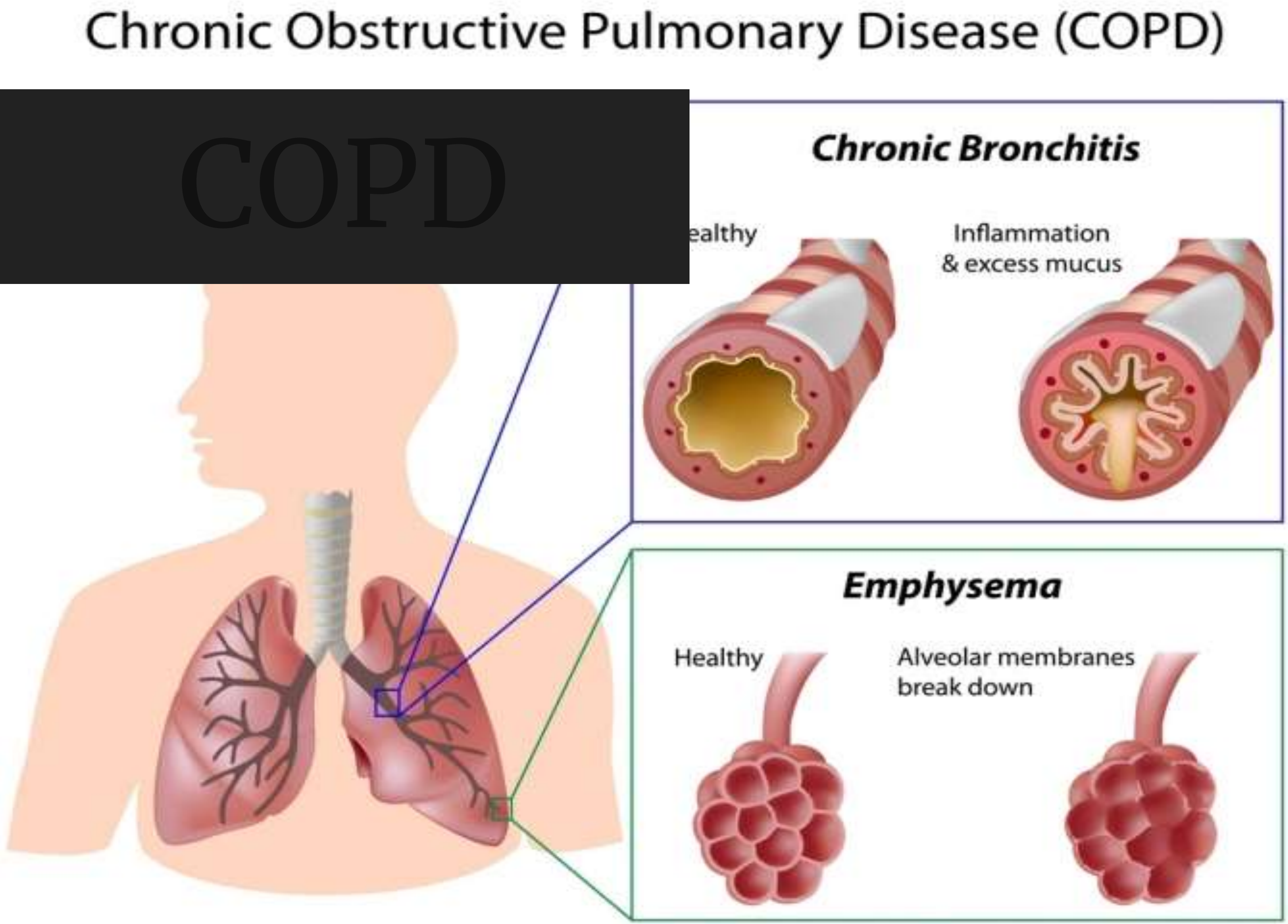
WATCH THESE VIDEOS

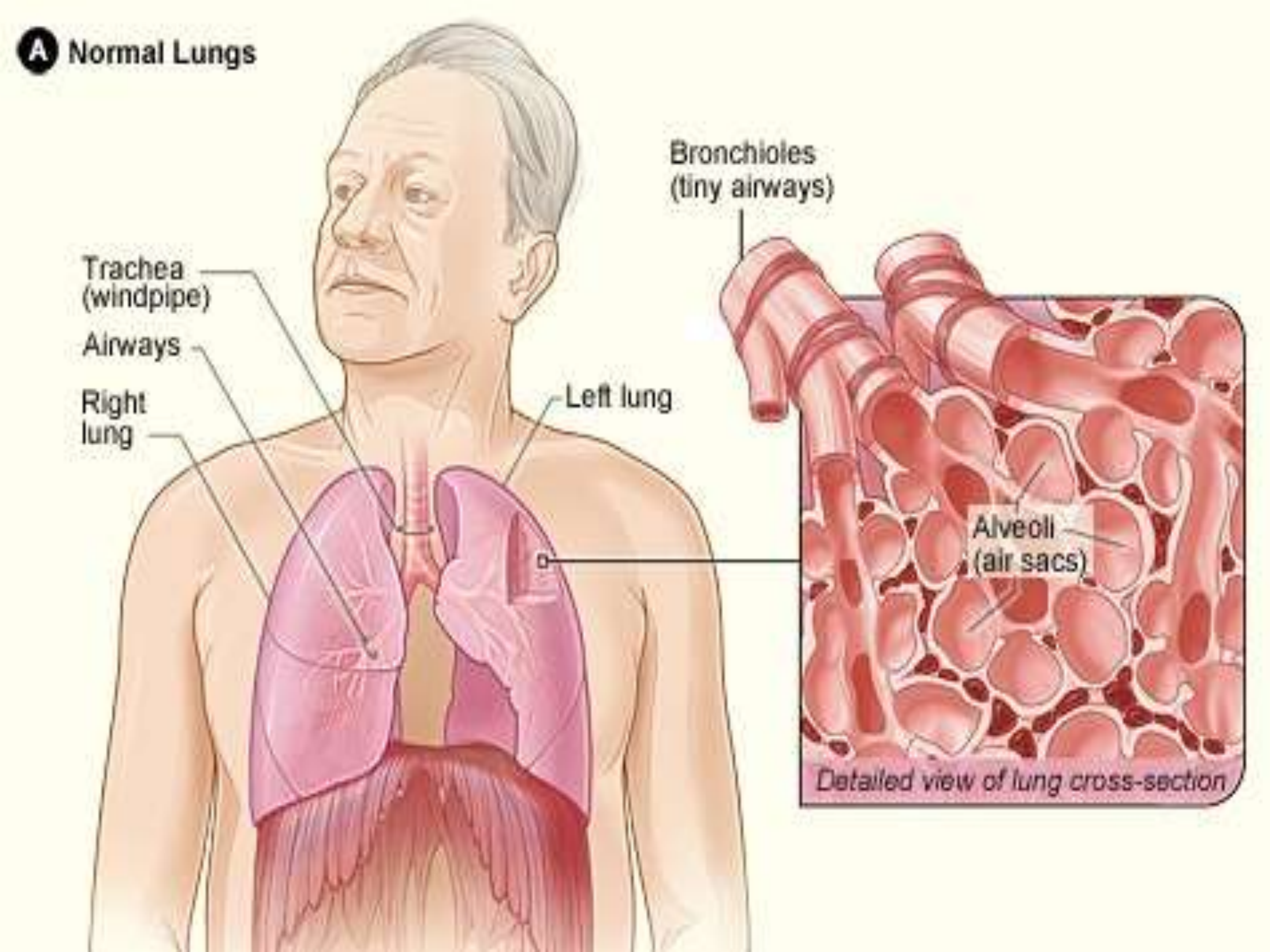
- **(Understanding COPD)** https://www.youtube.com/watch?v=T1G9Rl65M-Q
- **(What is COPD?)** https://www.youtube.com/watch?v=2nBPqSiLg5E
- **(Allergic rhinitis)** https://www.youtube.com/watch?v=3YkkNAvd9J4

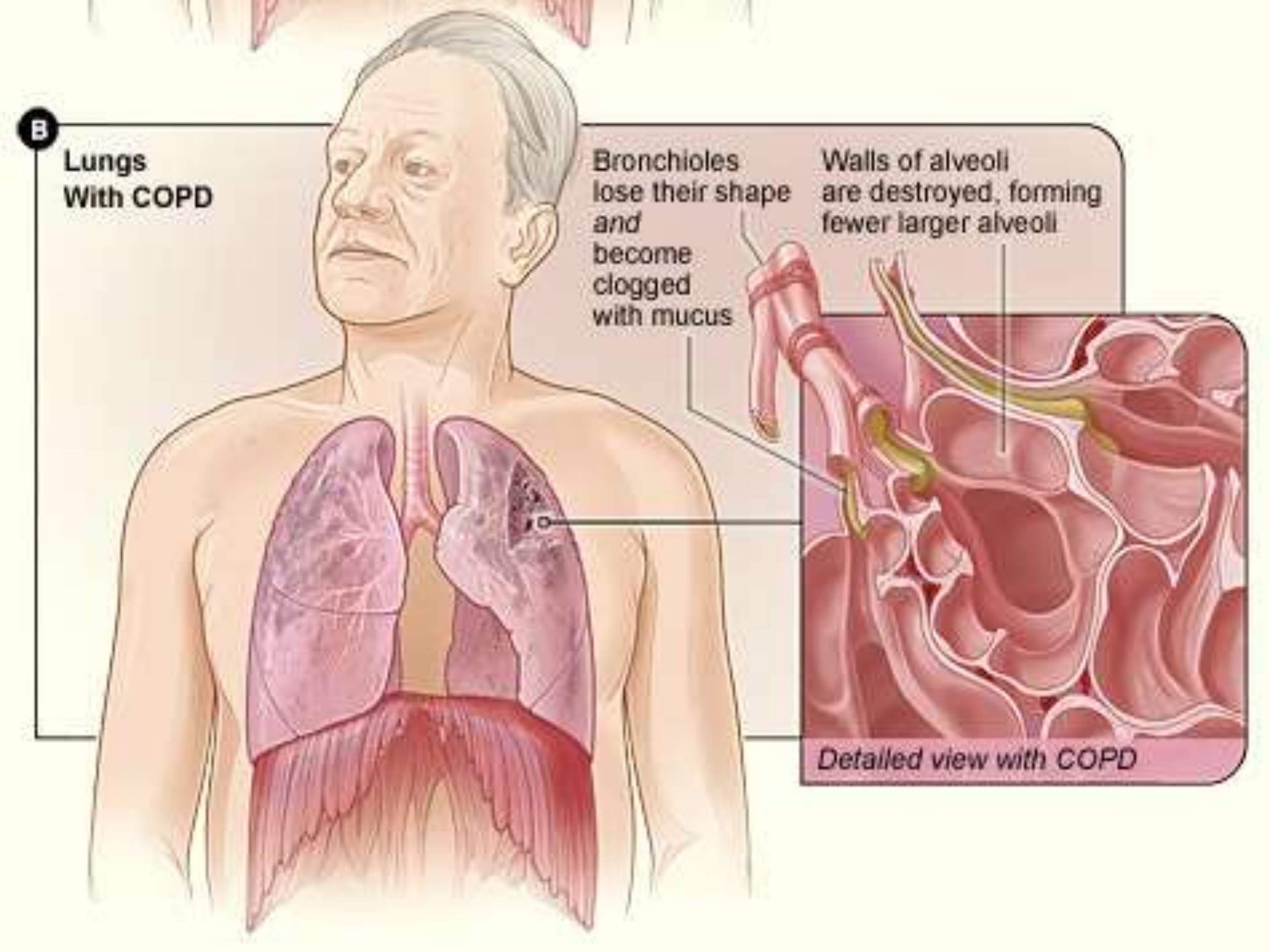
COPD

- Chronic obstructive pulmonary disease is a chronic, irreversible obstruction of airflow.
  - The airways and air sacs lose their elastic quality.
  - The walls between many of the air sacs are destroyed.
  - The walls of the airways become thick and inflamed.
  - The airways make more mucus than usual, which tends to clog them.

COPD


Chronic Obstructive Pulmonary Disease (COPD)
Chronic Bronchitis
ealthy
Inflammation
& excess mucus
Emphysema
Healthy
Alveolar membranes
break down


A Normal Lungs
Bronchioles
(tiny airways)
Trachea
(windpipe)
Airways
Right
lung
Left lung
Alveoli
(air sacs)
Detailed view of lung cross-section


B
Lungs
With COPD
Bronchioles
lose their shape
and
become
clogged
with mucus
Walls of alveoli
are destroyed, forming
fewer larger alveoli
Detailed view with COPD

COPD

DIAGNOSIS

SPIROMETRY

- measures how much air a person can inhale and exhale, and how fast air can move into and out of the lungs

COPD

Three of the major diseases that are grouped together as COPD:

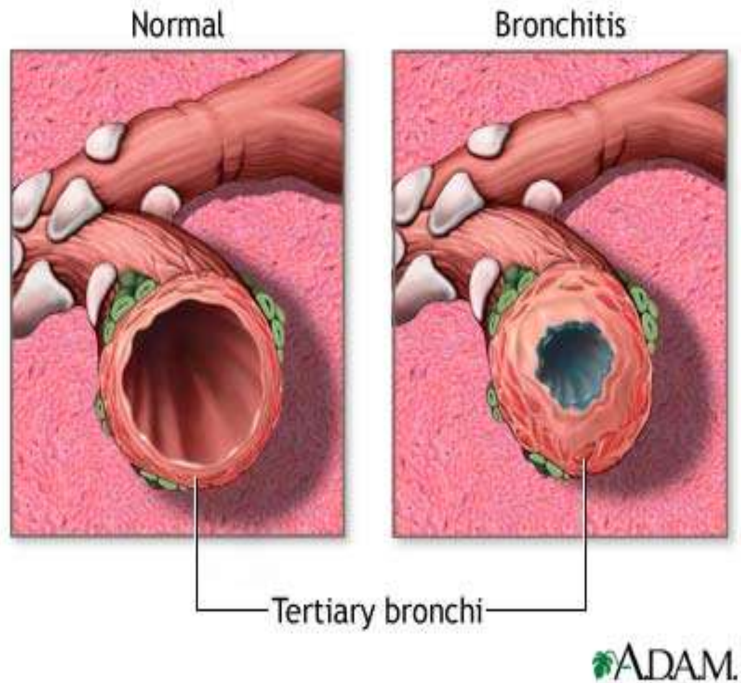
1. Asthmatic bronchitis
2. Chronic bronchitis
3. Emphysema

- Chronic Obstructive Lung Disease (COLD)
- Chronic Lower Respiratory

COPD

CHRONIC BRONCHITIS

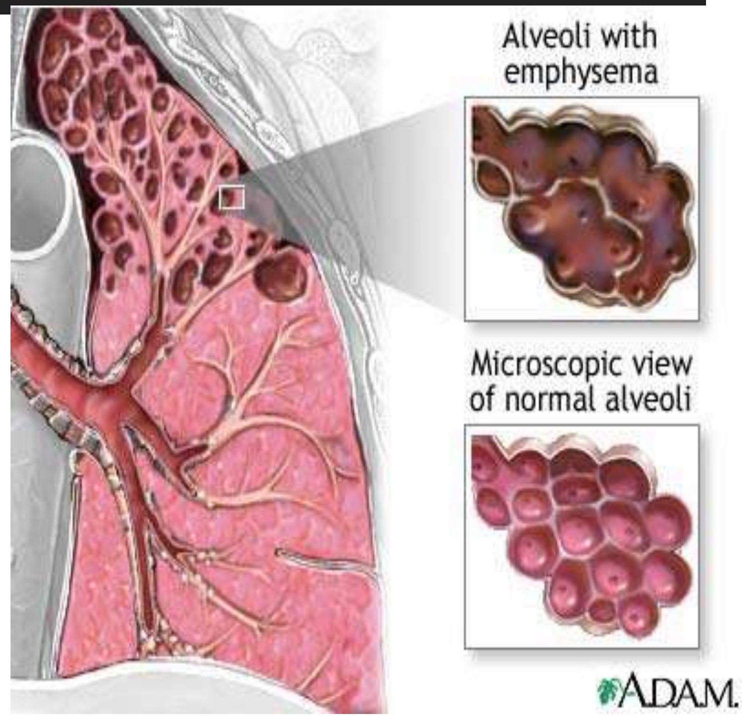
- characterized by
  - Chronic inflammation and excess mucus production
  - Presence of chronic productive cough



COPD

EMPHYSEMA

- characterized by
  - Damage to the small, sac-like units of the lung that deliver oxygen into the lung and remove the



MANAGEMENT COPD

MANAGEMENT

STAGES	RECOMMENDED TREATMENT
All stages	•Avoid risk factors such as smoking, irritants and allergens •Receive influenza vaccine annualy •Pneumococcal polysaccharide vaccine •Treat complications accordingly
STAGE 1: Mild COPD	Use short-acting bronchodilator as needed
STAGE 2: Moderate COPD	Maintenance therapy with 1 or more bronchodilators, pulmonary rehabilitation

MANAGEMENT COPD

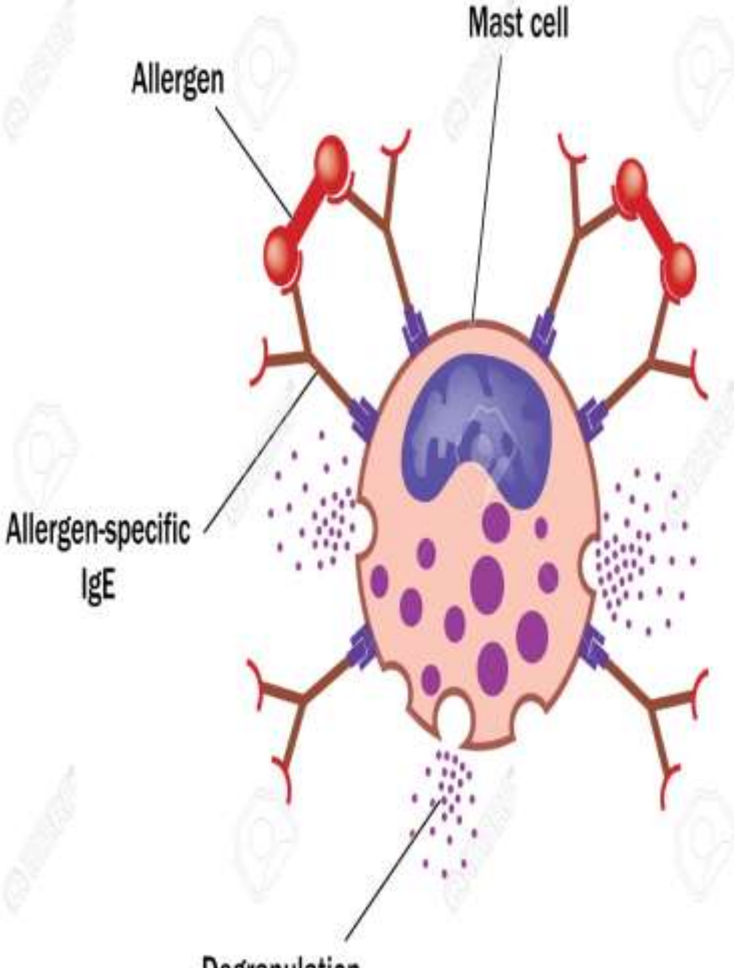
MANAGEMENT

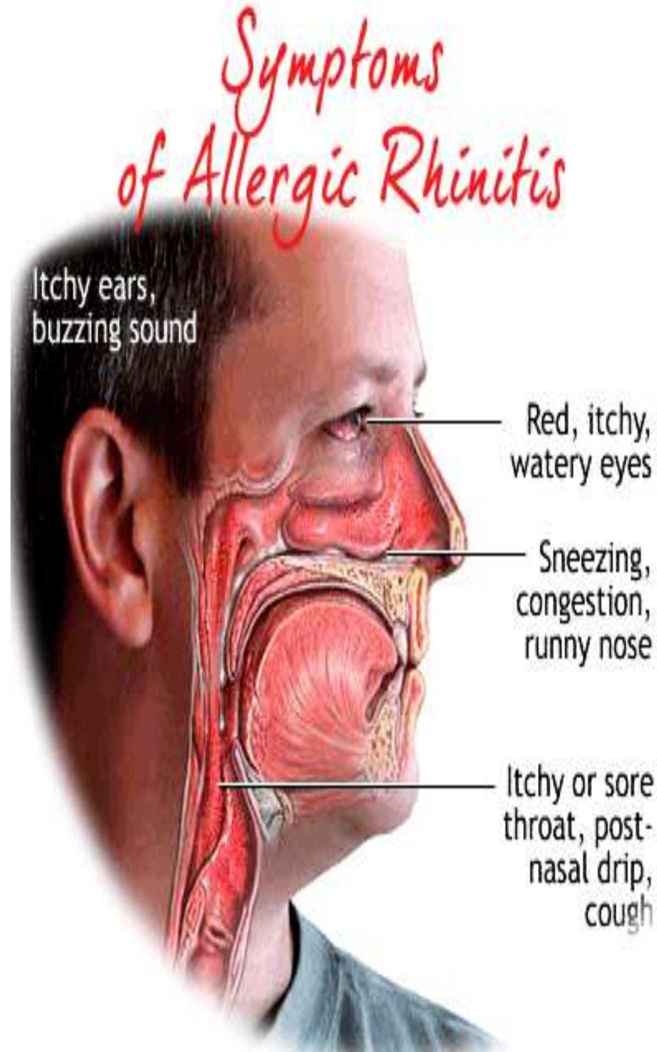
STAGES	RECOMMENDED TREATMENT
STAGE 3: Severe COPD	Maintenance therapy with 1 or more bronchodilators. Inhaled corticosteroids, pulmonary rehabilitation
STAGE 4: Very severe COPD	Regular treatment with 1 or more bronchodilators. Inhaled corticosteroids, pulmonary rehabilitation. Long-term oxygen therapy

ALLERGIC RHINITIS

ALLERGIC RHINITIS

- Rhinitis is an inflammation of the mucous membranes of the nose and is characterized by sneezing, itchy nose/eyes, watery rhinorrhea, and nasal congestion.
- An attack may be precipitated by inhalation of an allergen.
- The foreign material interacts with mast cells coated with IgE generated in response to a previous allergen exposure


Mast cell
Allergen
Allergen-specific IgE
Degranulation


Symptoms of Allergic Rhinitis
Itchy ears, buzzing sound
Red, itchy, watery eyes
Sneezing, congestion, runny nose
Itchy or sore throat, post-nasal drip, cough

ALLERGIC RHINITIS

TREATMENT

- Antihistamines
- α1-selective agonists
- Inhaled corticosteroids

ALLERGIC RHINITIS

TREATMENT

- Antihistamines
  - H1 receptor antagonists

1st Generation	2nd Generation
• More lipid soluble • More sedating	• Less lipid soluble • Less sedating

ALLERGIC RHINITIS

FIRST GENERATION		
CLASS	**DRUGS**	**BRAND NAME**
Ethanolamines	Carbinoxamine Dimenhydrinate Diphenhydramine	Clistin® Dramamine® Benadryl®
Piperazine	Hydroxyzine Cyclizine Meclizine	Atarax® Marezine® Bonine®
Alkylamines	Bropheniramine Chlorphrniramine	Dimetane® Chlor-Trimeton®
Phenothiazine	Promethazine	Phenergan®
Miscellaneous	Cyproheptadine	Periactin®

ALLERGIC RHINITIS

Antihistamines

SECOND GENERATION		
CLASS	**DRUGS**	**BRAND NAME**
Piperidine	Fexofenadine	Allegra®
Miscellaneous	Loratadine Desloratadine Cetirizine	Claritin® Clarinex® Zyrtec®

ALLERGIC RHINITIS

TREATMENT

- α1-selective agonists
  - Nasal decongestants

MOA:

Constrict dilated arterioles in the nasal mucosa and reduce airway resistance by binding at α1 receptors.

ALLERGIC RHINITIS

TREATMENT

- α1-selective agonists
  - Nasal decongestants

DRUGS:

- Phenylephrine
- Propylhexedrine
- Tetrahydrazoline
- Methoxamine
- Oxymetazoline

ALLERGIC RHINITIS

TREATMENT

- α1-selective agonists
  - Nasal decongestants

  S/E:

  Rhinitis medicamentosa
    - Oral – not be given >5 days
    - Spray – not be given >3 days

ALLERGIC RHINITIS

TREATMENT

- Inhaled corticosteroids

S/E: nasal irritation

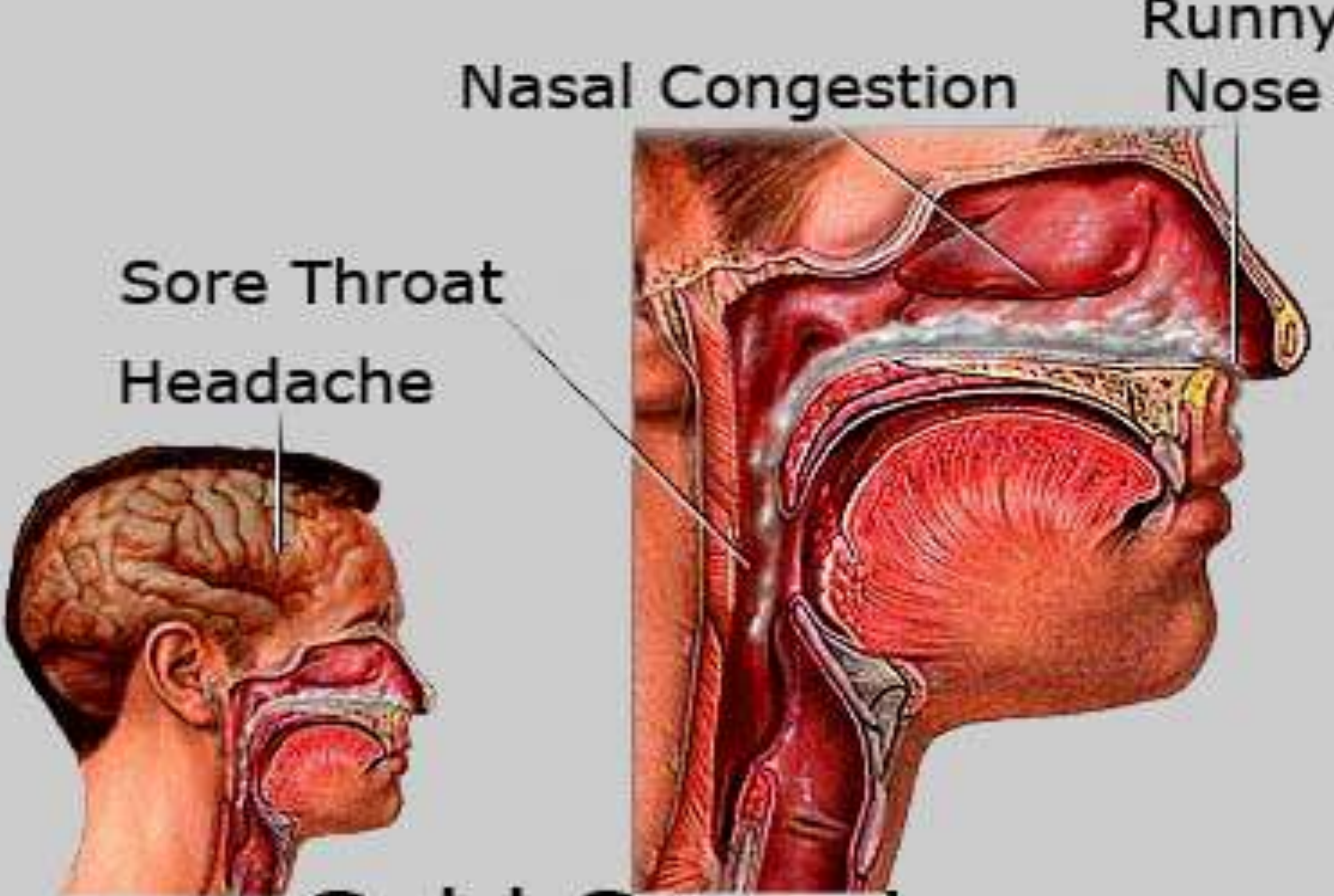
dryness

epistaxis

COUGH & COLDS

COUGH & COLDS

- Common colds - Virus invade the mucosa of the upper respiratory tract, nose, pharynx and larynx which leads to the upper respiratory system
- <u>Signs and symptoms</u>: excessive mucous production leads to sore throat, coughing, upset stomach.



Cold Symptoms

COUGH & COLDS

TREATMENTS

- Decongestants
  - vasoconstriction resulting in mucosal drying
- Antihistamines
  - combat increased histamine = nasal congestion and mucosal irritation
- Mucous removal
  - Mucolytic and expectorant
- Antitussives
  - work to suppress coughing

COUGH & COLDS

TREATMENTS

- **MUCOUS REMOVAL**
  - Aid in the coughing up and spitting out of the excess mucous that has accumulated in the respiratory tract by breaking down and thinning the secretions
  - ACTIONS:
    - MUCOLYTIC - Loosening and thinning the respiratory tract secretions
    - EXPECTORANT - Direct stimulation of the secretory glands in the respiratory tract.

COUGH & COLDS

TREATMENTS

- **MUCOUS REMOVAL**
  - MUCOLYTIC

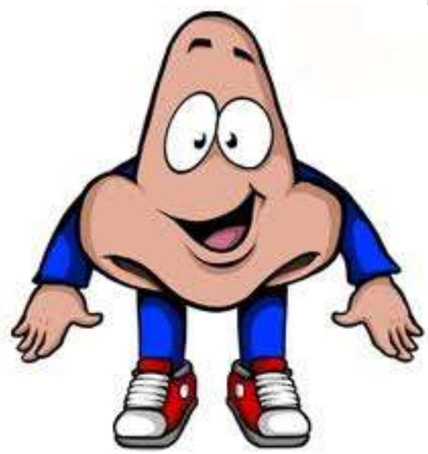
    Carbocisteine (Mucodyne®, Solmux®)
    Erdosteine (Erdotin®)
    Mecysteine (Visclair®)
    Acetylcisteine (Fluimicil®)
    Ambroxol (Mucosolvan®, Sinecod®)
    Bromhexine (Bisolvon®)

COUGH & COLDS

TREATMENTS

- **MUCOUS REMOVAL**
  - EXPECTORANT

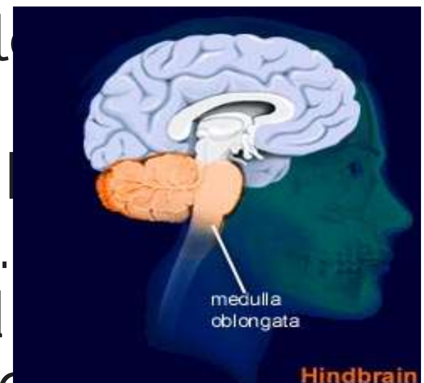
    Guaifenesin (Robitussin®



COUGH & COLDS

TREATMENTS

- **ANTITUSSIVE**
  - suppress the cough.
    - Use a central or a l
      mechanism.
  - Used to inhibit a coug
    via a central mechanism.
  - Cough center located
    in the medulla is targeted.



COUGH & COLDS

TREATMENTS

- **ANTITUSSIVE**
  - Opioid drugs all have antitussive effects
    - Codeine is the only opioid used as a cough medicine
    - gold-standard treatment for cough suppression
    - S/E: CNS and respiratory depression and addictive

COUGH & COLDS

TREATMENTS

- **ANTITUSSIVE**
  - Non opioid
    - Dextromethorphan
    - synthetic derivative of morphine
    - Safe, non-addicting and does not cause CNS or respiratory depression.

COUGH & COLDS

TREATMENTS

- **COLD PREPARATIONS**
  - **ROBITUSSIN DM®:** Dextromethorphan hydrobromide + guaifenesin
  - **TUSERAN FORTE® (REFORMULATED):** Dextromethorphan hydrobromide + phenylephrine hydrochloride + paracetamol

Thank you!

Any questions?

You can find me at:

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username@fatima.edu.ph

#RisetotheTop